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AI Engineering Internship-Weekly Report

WEEK 1 LEARNING REPORT

- **What I learned:**

Week 1 gave me a clearer picture of what happens behind an application when a user sends a request. I revised the Python basics first, including functions, classes and objects, inheritance, common collection types, and exception handling. One useful point for me was understanding why these are important in backend work rather than treating them as separate Python topics. Functions keep repeated logic in one place, classes help organize related data and behavior, and exception handling gives the program a controlled way to deal with problems.

- **Backend, HTTP, and REST:**

This part helps me connect python with actual backend development. I learn how the client and server communicate, and how HTTP provides the rules for that communication. I also became more comfortable with the main HTTP methods. The status codes were another important part of this week. The main thing is, I learned the particular differences between errors such as 401, 403, 404 and 422.

- **Fast API and Data Validation:**

Fast API was the main framework introduced in this handout. I learned how routes are defined, how path and query parameters were used, and how request and response models describe the data an API accepts and returns. In Pydantic validation it checks the wrong data before the main code runs. I also learned that Fast API has API documentation in /docs and /redoc. This helps me to test the API.

- **Databases:**

The database section showed me why an API cannot rely on python variables for storing information. Data in a database stays safe when the application is restarted. I learned about tables, primary keys, foreign keys, SQL queries and how tables are connected with each other. I also learned about SQLAlchemy and how it connects FASTAPI with a database.

- **My Main Takeaway:**

My biggest learning from week 1 is that backend development is not only about making an API that gives data. I learned that an API needs proper request and response flow,

correct HTTP and status codes, data validation, documentation, and a database to store data. I also learned why we use environment's and requirement.txt to setup a project. Overall, week 1 showed me how simple python can be used to make a real API.